

Highlights

MIRAUS: MRI-Informed Residual Adaptation for TRUS Prostate Boundary Measurement

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- Magnetic resonance images guide training for ultrasound prostate measurement.
- Neighboring magnetic resonance slices provide additional anatomical information.
- Training uses magnetic resonance and ultrasound; prediction uses ultrasound alone.
- Prostate size measurement supports both automatic and box-guided modes.
- The model improves prostate boundary accuracy on ultrasound images.

MIRAUS: MRI-Informed Residual Adaptation for TRUS Prostate Boundary Measurement

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ABSTRACT

Accurate prostate boundary measurement from transrectal ultrasound (TRUS) supports caliper-based size measurement, volume estimation, and downstream prostate-specific antigen density assessment, but speckle, shadowing, weak contrast, and deformation can impair accuracy. Magnetic resonance imaging (MRI) provides clearer anatomy but is impractical as a routine TRUS input. We propose MIRAUS, an MRI-informed residual adaptation framework that uses MRI as privileged information during training. To accommodate coarse MRI-TRUS correspondence, MIRAUS aggregates a local 2.5D MRI neighborhood with utility supervision to identify boundary-relevant anatomical context. The resulting teacher feature is decomposed into the original TRUS representation and an MRI-conditioned residual, which a lightweight residual adapter learns to predict from TRUS alone. The trained student model supports automatic and box-assisted boundary measurement without the input of MRI during the deployment. In patient-disjoint five-fold cross-validation on the μ RegPro dataset, automatic segmentation achieved 93.20% Dice and 3.62 mm HD95, reducing HD95 by 59.1% compared with nnU-Net. In a simulated clinical workflow using the conventional ellipsoid formula for prostate volume measurement, volumes calculated from model-measured diameters differed from those calculated from ground-truth diameters by a mean absolute difference of 4.68 mL. On a separate-center cohort, standardized box-assisted inference achieved 84.72% Dice, the highest among the evaluated box-assisted methods. MIRAUS brings training-time MRI guidance into routine TRUS measurement while improving boundary and caliper estimation and supporting flexible operator guidance.

1. Introduction

Prostate cancer represents a major global health burden as the second most frequently diagnosed malignancy in men, accounting for approximately 1.47 million new cases and 397 000 deaths worldwide in 2022 [1]. Given that advanced-stage disease remains largely incurable [2], the early detection and long-term surveillance are paramount, i.e. active screening has been associated with up to a 48.9% reduction in metastatic diagnosis risk [3]. In current urological practice, PSA testing is widely used for initial screening, but its limited specificity contributes to overdiagnosis and unnecessary invasive biopsies [4–6]. Because serum PSA levels are strongly confounded by gland size, prostate volume assessment shows great potential to calculate PSA density (PSAD), which incorporates gland volume into PSA interpretation and can improve pre-biopsy risk stratification [7].

Transrectal ultrasound (TRUS) is widely used for prostate assessment due to its real-time, portable, and inexpensive nature. In conventional examination, operators select representative axial and sagittal views and place transverse, antero-posterior, and longitudinal calipers. These three diameters are combined using a prolate-ellipsoid model to estimate the prostate volume [8, 9]. The transverse and anteroposterior

measurements depend directly on the perceived gland boundary, so contour errors propagate into the derived volume and PSAD. Speckle noise, acoustic shadowing, weak soft-tissue contrast, probe-induced deformation, view selection, and caliper placement therefore constitute major sources of measurement variation.

Automated TRUS segmentation has advanced through convolutional networks, spatial-context modeling, domain adaptation, and promptable medical foundation models [10, 11]. These approaches improve contour estimation, interaction, and computational efficiency, but the deployed prediction remains constrained by the anatomical evidence available in the TRUS input. Spatial-context methods can incorporate neighboring frames or complete 3D sweeps to provide richer contextual information, but such strategies require additional TRUS inputs beyond the selected measurement view. A complementary strategy is to introduce richer anatomical context during training, rather than requiring additional TRUS observations at deployment.

Magnetic resonance imaging (MRI) provides higher soft-tissue contrast than TRUS and can supply complementary anatomical information for prostate boundary estimation [12]. Multi-observer analyses have also demonstrated close structural correspondence between prostate delineations obtained from MRI and 3D TRUS [13]. MRI is nevertheless unsuitable as a routine input for rapid TRUS measurement because of its cost, acquisition time, and infrastructure requirements [14]. TRUS and MRI therefore offer complementary advantages: TRUS is suitable for routine, efficient examination, whereas

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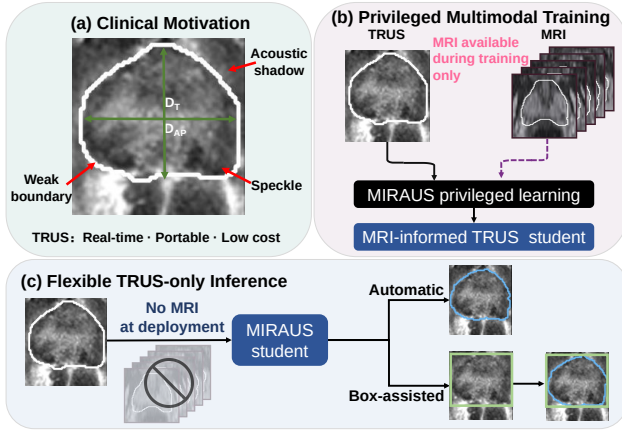


Figure 1: Clinical motivation and deployment concept of MIRAUS. (a) TRUS supports real-time prostate boundary and caliper measurement, but speckle, shadowing, and weak tissue contrast obscure the gland boundary. (b) Weakly aligned MRI provides privileged anatomical information during training. (c) The deployed student model performs automatic or box-assisted boundary measurement from a single TRUS frame without MRI.

MRI provides clearer anatomical context. This complementary relationship motivates a setting based on the learning using privileged information (LUPI) paradigm, in which MRI is utilized only during the training stage. In other words, the deployed system can operate on TRUS alone [15].

This formulation, however, will further involve the challenge of cross-modal transfer. MRI and TRUS are acquired independently, and differences in patient positioning, acquisition orientation, slice spacing, and probe pressure limit their spatial correspondence [16]. Their anatomical relationship therefore generally remains only at a coarse level, making direct cross-modal supervision or dense feature alignment unreliable. Existing multimodal and missing-modality approaches provide another route to richer representations, but they commonly assume spatially aligned modalities or transfer cohort-level multimodal information. Fully unpaired transfer avoids the correspondence requirement, but it also discards the coarse within-patient anatomical relationship available in paired acquisitions. This setting creates two coupled challenges. First, useful anatomical context must be identified from neighboring MRI slices despite only coarse MRI-TRUS correspondence. Second, the additional information contributed by MRI must be isolated and transferred to a student model that ultimately operates from TRUS alone.

To address these challenges, we propose MIRAUS, an MRI-informed residual adaptation framework for TRUS prostate boundary measurement. Rather than assuming direct slice-to-slice registration, coarse relative-depth correspondence defines a local 2.5D MRI neighborhood for each TRUS frame, and utility-supervised aggregation emphasizes the candidates that provide useful boundary information. We then decompose the MRI-conditioned teacher feature into the original TRUS representation and an additional feature

increment defined by their difference. This residual captures the correction contributed by privileged MRI beyond what is already encoded from TRUS. A lightweight adapter learns to predict this increment from TRUS alone and adds it back to the TRUS feature for boundary estimation. At deployment, the MRI branch and teacher model are removed. Mixed-prompt adaptation allows the same student model to operate automatically with a full-image prompt or interactively with a coarse localization box, providing flexible measurement modes without additional imaging input.

The main contributions of this work are summarized as follows¹:

- **Local multi-slice MRI context with utility-guided aggregation:** We exploit a relative-depth-guided 2.5D MRI neighborhood to draw complementary information from adjacent slices, while boundary-utility supervision prioritizes the candidates most useful for the current TRUS frame.
- **MRI-informed residual adaptation:** We represent the privileged teacher feature as the original TRUS representation plus an MRI-conditioned feature increment, and train a lightweight residual adapter to recover this correction from TRUS alone for MRI-free deployment.
- **Dual-mode TRUS measurement and deployment evaluation:** The same TRUS-only student model supports automatic and box-assisted boundary measurement, from which in-plane calipers and conventional ellipsoid volumes are derived. We further assess boundary accuracy, operator localization, interaction time, inference latency, anatomical-depth sensitivity, and cross-center performance.

2. Related Work

2.1. TRUS Prostate Boundary and Volume Measurement

Conventional TRUS prostate volumetry derives gland volume from transverse, anteroposterior, and longitudinal diameters acquired on selected axial and sagittal views. The diameters are combined using an ellipsoid approximation, providing a rapid measurement that is compatible with routine ultrasound examination [17]. Its reliability depends on the selected imaging plane, the visibility of the gland margin, and the placement of each caliper. Observer studies have consequently reported variation in both TRUS boundary delineation and derived prostate volume [18]. The ellipsoid model also introduces a geometric error when the gland departs from the assumed shape, which can alter the resulting volume and PSAD.

Boundary estimation is therefore an integral part of the measurement process. Early automated approaches combined handcrafted appearance descriptors, statistical shape

¹Source code and an interactive case explorer are publicly available at <https://github.com/Zeming-Huang/MIRAUS> and <https://zeming-huang.github.io/MIRAUS/>.

constraints, deformable contours, and level-set optimization to stabilize prostate margins under speckle and acoustic attenuation [19, 20]. These methods introduced anatomical regularization into TRUS boundary extraction, but their performance depended on manually designed features and shape assumptions. Their development established the value of boundary geometry for reproducible measurement while motivating data-driven models that can learn more complex TRUS appearances.

2.2. Automated and Promptable TRUS Boundary Estimation

Convolutional networks shifted TRUS boundary estimation from engineered descriptors to learned image representations. Slice-wise models support rapid processing during ultrasound-guided examination, whereas 3D architectures use complete sweeps to model spatial continuity [21–24]. Neighboring-slice input has also been investigated as an intermediate strategy, although additional TRUS slices do not consistently improve segmentation [25]. TRUS-specific architectures further incorporate attention, scanner adaptation, shadow modeling, and domain-aware augmentation [26–30]. These methods improve boundary estimation across acquisition conditions, but spatial-context models commonly require neighboring frames or complete volumes at inference.

Promptable foundation models provide a complementary interaction mechanism. The Segment Anything Model introduced point-, box-, and mask-conditioned segmentation, and MedSAM adapted this interface to medical images [11, 31]. SAMed and SAM-Med2D support automatic or prompt-assisted medical segmentation, while SAMUS introduces ultrasound-specific local representations [32–34]. Lightweight transformer encoders reduce the computational cost of promptable inference, and volumetric systems such as SegVol and nnInteractive extend interaction to 3D data [35–37]. These developments improve interaction and deployment efficiency, but they still estimate the boundary from the anatomical information available in the deployed TRUS input. Training-time access to a clearer modality therefore offers a complementary route to improve TRUS boundary measurement without increasing examination-time imaging requirements.

2.3. Privileged Learning with Weakly Aligned Multimodal Data

Knowledge distillation transfers information from a teacher model to a deployable student model through output distributions or intermediate representations. Generalized distillation connects this formulation with LUPI, in which auxiliary information is available during training but absent during inference [15, 38]. Modality hallucination similarly trains a single-modality network to reproduce representations learned from a richer training-time input [39].

Medical imaging has applied privileged learning to uncertainty estimation, cross-modal feature supervision, and single-modality deployment [40, 41]. Privileged Modality Distillation transfers complementary information between

intravascular imaging modalities, while KD-Net distills a multimodal brain-tumor teacher model into a monomodal student model [42, 43]. Missing-modality architectures such as RFNet and mmFormer learn multimodal representations that remain usable when an input modality is unavailable, and ProtoKD transfers prototype-level knowledge to a single-modality model [44–46]. These formulations are primarily designed for modalities with dependable spatial correspondence.

MRI guidance for prostate TRUS analysis has also been introduced through deformable registration and MRI-derived anatomical priors. Dense MRI–TRUS correspondence remains difficult because patient positioning, acquisition orientation, slice spacing, and probe pressure alter prostate geometry and image appearance [47]. Direct feature matching under these conditions may associate the current TRUS frame with irrelevant MRI anatomy. Fully unpaired transfer avoids this correspondence requirement, but it discards the coarse within-patient anatomical relationship available in paired acquisitions [48].

Together, these studies show that anatomical priors can stabilize prostate boundaries, multimodal distillation can transfer information available only during training, and promptable models can support different levels of operator guidance. The remaining challenge is to transfer useful MRI anatomy under coarse MRI–TRUS correspondence while preserving TRUS-only deployment. MIRAUS addresses this challenge by aggregating local MRI features through relative-depth correspondence and distilling the resulting MRI-conditioned residual into a TRUS-only student model for automatic and box-assisted boundary measurement.

3. Methods

3.1. Measurement Task and Measurement Model

The primary measurement output of MIRAUS is the two-dimensional prostate boundary on a selected TRUS frame. Let $\mathbf{x}_n \in \mathbb{R}^{H \times W}$ denote a single TRUS frame and let $\mathbf{b}$ denote either a full-image prompt or an optional coarse localization box. The deployed student model produces a prostate probability map $\hat{\mathbf{p}}_n$, from which the binary mask $\hat{\mathbf{y}}_n$ and its boundary $\hat{\mathcal{C}}_n = \partial \hat{\mathbf{y}}_n$ are obtained. The boundary coordinates are initially expressed on the image grid and are converted into physical coordinates using the corresponding pixel spacing.

Two in-plane linear measurements are derived from $\hat{\mathcal{C}}_n$. The left–right diameter $\hat{D}_T$ is the transverse extent of the estimated boundary, and the anteroposterior diameter $\hat{D}_{AP}$ is its orthogonal axial extent:

$$\begin{aligned}\hat{D}_T &= \max_{(x,y) \in \hat{\mathcal{C}}_n} (x) - \min_{(x,y) \in \hat{\mathcal{C}}_n} (x), \\ \hat{D}_{AP} &= \max_{(x,y) \in \hat{\mathcal{C}}_n} (y) - \min_{(x,y) \in \hat{\mathcal{C}}_n} (y).\end{aligned}\tag{1}$$

where x and y denote physical coordinates in millimetres. The corresponding reference quantities, D_T^{ref} and D_{AP}^{ref} , are

derived from the expert reference mask $\mathbf{y}_n$ using the same procedure.

Conventional prostate volume measurement approximates the gland as a prolate ellipsoid [8]:

$$V_{\text{ellip}} = \frac{\pi}{6} D_T D_{AP} D_{SI}, \quad (2)$$

where D_{SI} is the superior–inferior diameter. Diameters are measured in millimetres, and the resulting volume is converted from mm^3 to millilitres.

The retrospective caliper experiment evaluates replacement of the two in-plane measurements on a reference-selected maximal axial cross-section. The model supplies $\hat{D}_T$ and $\hat{D}_{AP}$, while the reference longitudinal diameter D_{SI}^{ref} is held fixed. The resulting assisted estimate is

$$\hat{V}_{\text{ellip}} = \frac{\pi}{6} \hat{D}_T \hat{D}_{AP} D_{SI}^{\text{ref}}. \quad (3)$$

This protocol measures the error introduced when model-derived boundaries replace the two axial calipers within the conventional ellipsoid workflow.

3.2. MIRAUS Framework

We formulate TRUS prostate boundary measurement under the Learning Using Privileged Information (LUPI) paradigm [15]. Each training sample contains a TRUS frame $\mathbf{x}_n$, its reference mask $\mathbf{y}_n$, and a weakly aligned local MRI neighborhood $\mathcal{M}_n$:

$$D_{\text{tr}} = \{(\mathbf{x}_n, \mathcal{M}_n, \mathbf{y}_n)\}_{n=1}^N. \quad (4)$$

MRI is used by the privileged teacher model during training, while the deployed model receives only the current TRUS frame and its prompt.

The overall privileged-learning and deployment pathways are illustrated in Fig. 2. First, the privileged teacher model selects and aggregates useful anatomical information from the local MRI neighborhood. Second, a lightweight residual adapter learns to recover the resulting MRI-conditioned feature increment from TRUS alone. Finally, mixed-prompt adaptation prepares the same student model for automatic full-image operation and box-assisted measurement. The MRI branch and teacher model are removed before deployment.

3.2.1. Modality-Specific Encoding and Local MRI Context

Densely annotated TRUS datasets remain limited, making it difficult to train a high-capacity segmentation model from scratch. Ultrasound examination also favors a compact model that can process individual frames efficiently. We therefore build MIRAUS upon the LiteMedSAM architecture, which contains a TinyViT image encoder, a lightweight prompt encoder, and a mask decoder. The four-stage TinyViT backbone converts each 256×256 input into a 256-channel feature map with a spatial resolution of 64×64 .

TRUS and MRI differ substantially in appearance, so they are processed by separate image encoders. We use two TinyViT encoders with the same architecture but independent

modality-specific pretrained weights. Let E_U and E_M denote the TRUS and MRI encoders, respectively. Given a TRUS frame $\mathbf{x}_n$ and an MRI candidate $\mathbf{m}_n^j$, they produce

$$\mathbf{F}_U = E_U(\mathbf{x}_n), \quad \mathbf{F}_M^j = E_M(\mathbf{m}_n^j). \quad (5)$$

Both image encoders remain frozen during teacher training and student distillation. Training is therefore concentrated on the cross-modal interaction, aggregation, residual adaptation, and mask-prediction modules.

A single MRI slice captures only one cross-section of the prostate. Adjacent slices provide complementary information about gland shape and its gradual variation along the acquisition direction. We therefore extend the privileged MRI context from 2D to 2.5D by constructing a local five-slice neighborhood for each TRUS frame.

We first assign an anchor MRI slice q_n using

$$s_n = \frac{r_n}{R_n - 1}, \quad q_n = \text{round}(s_n(L_n - 1)), \quad (6)$$

where r_n is the TRUS slice index, R_n is the number of TRUS slices, and L_n is the number of MRI slices. The resulting neighborhood is $\mathcal{M}_n = \{(\mathbf{m}_n^j, v_n^j)\}_{j=-2}^2$, where $v_n^j \in \{0, 1\}$ marks whether the candidate lies within the available MRI range. The relative-depth mapping defines a coarse local search region, within which the utility selector identifies the MRI candidates that contribute to the teacher representation.

3.2.2. Utility-Supervised Multi-Slice MRI Aggregation

The local MRI neighborhood provides richer anatomical context, but its candidates are not equally useful for the current TRUS frame. MIRAUS therefore processes each candidate separately and learns a soft aggregation over the five MRI slices. The aggregation considers TRUS–MRI feature compatibility, relative slice position, and the candidate’s contribution to prostate boundary estimation.

Each MRI candidate first interacts with the TRUS feature through asymmetric multi-head cross-attention. TRUS supplies the query, whereas MRI supplies the keys and values:

$$\mathbf{A}_j = \text{LN}_2 \left[\mathbf{F}_U + W_O \text{MHA} \left(W_Q \text{LN}_1(\mathbf{F}_U), W_K \text{LN}_1(\mathbf{F}_M^j), W_V \text{LN}_1(\mathbf{F}_M^j) \right) \right]. \quad (7)$$

This interaction keeps the current TRUS frame as the representation anchor while allowing each MRI candidate to provide complementary anatomical information.

The five interacted candidates must then be weighted. Learnable mappings first generate normalized feature descriptors for the TRUS frame and each MRI slice:

$$\mathbf{d}_U = \frac{g_U [\rho(\mathbf{F}_U, \mathbf{b}_\Omega)]}{\|g_U [\rho(\mathbf{F}_U, \mathbf{b}_\Omega)]\|_2}, \quad \mathbf{d}_M^j = \frac{g_M [\rho(\mathbf{F}_M^j, \mathbf{b}_\Omega)]}{\|g_M [\rho(\mathbf{F}_M^j, \mathbf{b}_\Omega)]\|_2}. \quad (8)$$

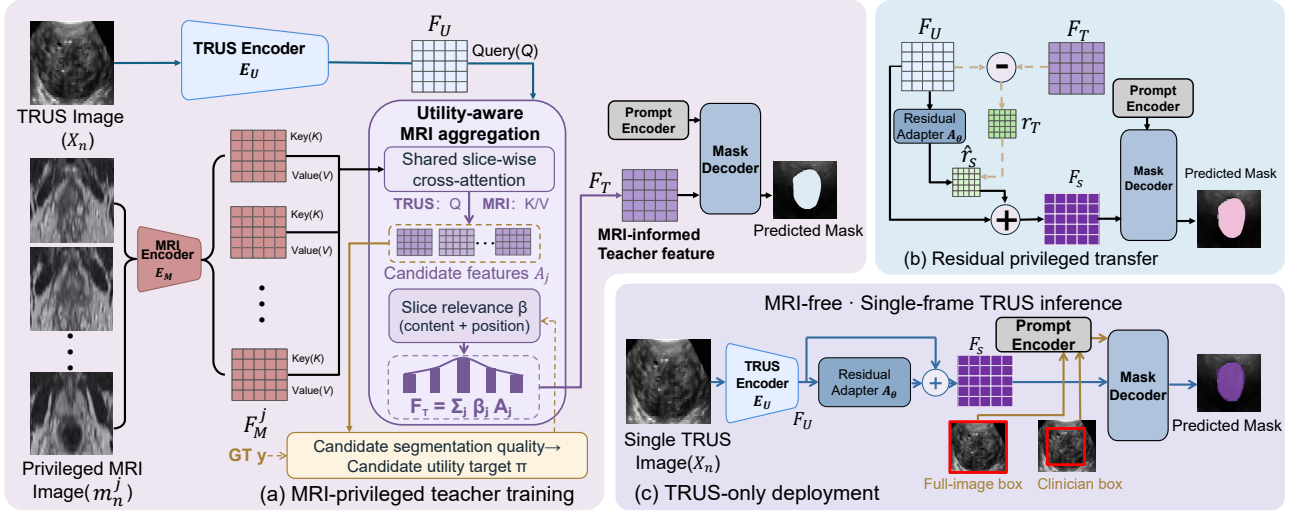


Figure 2: Overview of MIRAUS. (a) During MRI-privileged teacher model training, separate encoders extract the TRUS feature F_U and candidate MRI features F_M^j . Each MRI candidate interacts with F_U through shared slice-wise cross-attention. Candidate boundary-estimation quality provides a utility target for the slice weights β , and their weighted aggregation produces the MRI-informed teacher feature F_T . (b) During residual privileged transfer, the frozen teacher model defines the MRI-conditioned increment $r_T = F_T - F_U$. The student residual adapter predicts $\hat{r}_S$ from F_U alone and adds it to the TRUS feature to obtain F_S for boundary estimation. (c) At deployment, the MRI encoder and teacher model are removed. The same single-frame TRUS student model accepts either a full-image box for automatic measurement or a clinician-provided box for assisted measurement.

Here, $\rho(\cdot)$ concatenates global, prompt-region, and prompt-boundary pooled features, while g_U and g_M are modality-specific projections. During teacher training, $\mathbf{b}_\Omega$ spans the full image.

The content compatibility between the current TRUS frame and MRI candidate j is measured by cosine similarity. The selector combines this similarity with a learned relative-position bias and a weak Gaussian prior:

$$a_j = \alpha \langle \mathbf{d}_U, \mathbf{d}_M^j \rangle + u_j + \lambda_p \log p_j, \quad (9)$$

where $\alpha > 0$ controls the contribution of content similarity, u_j is a learned relative-position bias, and $p_j \propto \exp(-j^2/2\sigma_p^2)$. The spatial prior favors nearby candidates without forcing the assigned center to receive the largest weight.

Let $\mathcal{V}_n = \{j \mid v_n^j = 1\}$ denote the valid MRI candidates. Their normalized aggregation weights are

$$\beta_j = \begin{cases} \frac{\exp(a_j)}{\sum_{k \in \mathcal{V}_n} \exp(a_k)}, & j \in \mathcal{V}_n, \\ 0, & j \notin \mathcal{V}_n. \end{cases} \quad (10)$$

Feature compatibility alone cannot determine whether an MRI candidate improves the boundary estimate for the current TRUS frame. We therefore supervise the selector using candidate-specific boundary utility. A lightweight auxiliary head C_{cand} predicts a mask from each interacted feature $\mathbf{A}_j$, and the corresponding segmentation error is

$$e_j = \mathcal{L}_{\text{Dice}}(C_{\text{cand}}(\mathbf{A}_j), \mathbf{y}_n) + \mathcal{L}_{\text{BCE}}(C_{\text{cand}}(\mathbf{A}_j), \mathbf{y}_n). \quad (11)$$

A lower error indicates that the corresponding MRI candidate provides more useful information for estimating the current

prostate boundary. The errors are converted into the utility distribution

$$\pi_j = \begin{cases} \frac{\exp(-e_j/T_u)}{\sum_{k \in \mathcal{V}_n} \exp(-e_k/T_u)}, & j \in \mathcal{V}_n, \\ 0, & j \notin \mathcal{V}_n, \end{cases} \quad (12)$$

where T_u controls the sharpness of the target distribution. The utility target is detached before it supervises the predicted aggregation weights:

$$\mathcal{L}_{\text{util}} = D_{\text{KL}}(\text{sg}(\boldsymbol{\pi}) \parallel \boldsymbol{\beta}), \quad (13)$$

where $\text{sg}(\cdot)$ denotes the stop-gradient operator. This supervision teaches the selector to prioritize candidates according to their boundary-estimation utility, complementing the visual similarity and relative-position terms.

Finally, the valid candidate features are combined using the learned weights:

$$\mathbf{F}_T = \sum_{j \in \mathcal{V}_n} \beta_j \mathbf{A}_j. \quad (14)$$

The resulting teacher feature preserves the TRUS representation as its base while incorporating the MRI information selected for the current frame.

3.2.3. Residual Privileged Adaptation

After training, the complete teacher model is frozen. It generates target features for training the student model, with no gradients propagated into its components.

We express the teacher representation relative to the shared frozen TRUS feature:

$$\mathbf{r}_T = \text{sg}(\mathbf{F}_T - \mathbf{F}_U). \quad (15)$$

The student model uses a lightweight residual adapter A_θ to estimate this increment from $\mathbf{F}_U$ alone:

$$\hat{\mathbf{F}}_S = A_\theta(\mathbf{F}_U), \quad \mathbf{F}_S = \mathbf{F}_U + \hat{\mathbf{F}}_S. \quad (16)$$

The adapter contains two parallel convolutional branches whose outputs are summed. The final layer of one branch is zero-initialized to stabilize the beginning of training the student model.

Because both representations share the same frozen TRUS feature, SmoothL1 matching of the complete features is equivalent to residual matching under identical spatial averaging. This formulation makes the student-predicted correction explicit.

The predicted correction is aligned with the teacher-defined increment using

$$\mathcal{L}_{\text{priv}} = \text{SmoothL1}(\hat{\mathbf{r}}_S, \mathbf{r}_T). \quad (17)$$

The loss is averaged over all channels and spatial positions of the feature map.

Together with segmentation supervision, the complete student objective is

$$\mathcal{L}_S = \mathcal{L}_{\text{seg}} + 2\mathcal{L}_{\text{priv}}, \quad (18)$$

where $\mathcal{L}_{\text{seg}}$ combines Dice, binary cross-entropy, and mask-quality regression losses. Once training of the student model is complete, neither $\mathbf{F}_T$ nor $\mathbf{r}_T$ is required for inference.

3.3. Training Objectives and Mixed-Prompt Adaptation

Training comprises three sequential stages. First, the privileged teacher model receives the TRUS frame, its local MRI neighborhood, and the reference mask. Both image encoders remain frozen, while the cross-modal interaction, MRI aggregation, candidate head, and teacher-side segmentation components are optimized. The candidate head receives direct segmentation supervision, and its detached candidate errors define the utility target π that supervises the aggregation weights β through $\mathcal{L}_{\text{util}}$.

Second, the trained teacher model is frozen and used only for forward computation. It supplies the residual target $\mathbf{r}_T$, while the student residual adapter predicts $\hat{\mathbf{r}}_S$ from $\mathbf{F}_U$. The student representation $\mathbf{F}_S$ is optimized using the inherited segmentation objective together with $2\mathcal{L}_{\text{priv}}$. No gradients are propagated into the teacher model or MRI encoder during this stage.

Third, mixed-prompt adaptation prepares the same student model for automatic and box-assisted measurement. During transrectal acquisition, the probe is positioned close to the gland, and the prostate commonly occupies a large proportion of the image. A full-image box therefore provides a loose localization prompt for automatic operation:

$$\mathbf{b}_\Omega = [0, 0, W - 1, H - 1]. \quad (19)$$

Image quality, boundary ambiguity, and the preferred level of operator control vary across examinations. Mixed-prompt adaptation therefore combines full-image rehearsal

with perturbed localization boxes $\mathbf{b}_n^{\text{loc}}$. This strategy retains the full-image behavior while teaching the prompt encoder and mask decoder to use coarse localized guidance. During this stage, the TRUS encoder and residual adapter remain frozen, while the prompt encoder and mask decoder are optimized using $\mathcal{L}_{\text{seg}}$.

At deployment, the prediction of the student model is

$$\hat{\mathbf{p}}_n = S_\theta(\mathbf{x}_n, \mathbf{b}), \quad \mathbf{b} \in \{\mathbf{b}_\Omega, \mathbf{b}_n^{\text{loc}}\}. \quad (20)$$

Both operating modes use the same TRUS student model. MRI and the privileged teacher model are absent from the deployed measurement path.

3.4. Experimental protocol

3.4.1. Datasets and Evaluation Cohorts

We evaluated MIRAUS across two distinct clinical cohorts to rigorously assess both volumetric accuracy and domain generalization. The internal cohort comprises the public μRegPro dataset [49], containing paired, unaligned MRI and TRUS acquisitions from 73 patients. Model performance was evaluated using patient-disjoint 5-fold cross-validation. Pooling out-of-fold predictions across all 73 patients yielded 3,885 gland-bearing TRUS frames for 3D patient-macro analysis. To evaluate direct cross-center transferability under real-world domain shifts, we further established an independent external test cohort consisting of 327 gland-bearing 2D TRUS frames collected at China–Japan Friendship Hospital. The collection and use of this external cohort were approved by the Science and Technology Ethics Committee of the University of Chinese Academy of Sciences (approval no. UCASSTEC26-003; approval date: 4 February 2026), with written informed consent obtained from all participants. This external dataset lacks paired MRI acquisitions and was used strictly to evaluate the deployment of the trained TRUS-only student model. Dataset statistics and evaluation settings are summarized in Table 1.

All models were implemented in PyTorch and trained on NVIDIA GPUs. For MIRAUS, TRUS and MRI inputs were resized to 256×256 . Both TRUS and MRI TinyViT backbones remained frozen throughout optimization to leverage foundation vision priors. Training proceeded in three sequential stages using the AdamW optimizer with an initial learning rate of 5.5×10^{-4} , weight decay of 8.1×10^{-5} , and cosine learning rate scheduling. In Stage 1, the privileged teacher model was trained for 20 epochs (512 sampled frames per epoch, batch size 2), aggregating multi-slice MRI candidates via utility-supervised KL divergence. In Stage 2, the teacher model was frozen, and the student model was distilled for 20 epochs using the objective in Eq. (18). In Stage 3, the student model underwent 5 epochs of mixed-prompt adaptation, during which prompts were randomly sampled from standard full-image prompts $\mathbf{b}_\Omega$ and noise-perturbed bounding boxes $\mathbf{b}_n^{\text{loc}}$.

3.4.2. Evaluation Metrics and Retrospective Volumetric Protocol

For the μRegPro cohort, metrics were calculated from predictions pooled within each patient and summarized

Table 1
Summary of Datasets and Evaluation Protocols

Dataset	Protocol	Frames	Modalities
μ RegPro	5-fold patient CV	3,885	TRUS + MRI (training)
China–Japan Friendship Hospital	External test	327	TRUS only

across all 73 patients with equal weighting. HD95 was measured in millimetres on the 0.8-mm isotropic grid and in pixels for the external cohort, which lacked validated pixel-spacing metadata. Standardized box-assisted comparisons used identical reference-mask-derived boxes across methods; we further conducted experiments with bounding boxes drawn by multiple clinical observers in Section 5.2. Segmentation performance was assessed via Dice, IoU and HD95, while volume error was quantified by MAE in millilitres. The retrospective caliper experiment followed Section 3.1, comparing model-assisted ellipsoid estimates with both reference-caliper ellipsoid volumes and voxel-count reference volumes. Reference volumes were computed from the ground-truth masks using the isotropic voxel spacing; the reference superior-inferior diameter was their full longitudinal extent.

4. Results

4.1. Main Benchmark Comparison and Cross-Site Generalization

On the public μ RegPro dataset, MIRAUS achieved the best performance among the evaluated baselines across 73 patients (3,885 gland-bearing TRUS frames). By leveraging privileged multi-slice MRI guidance during training, the network operates in a lightweight, TRUS-only manner at inference, avoiding MRI processing and cross-modal alignment at deployment.

As summarized in Table 2, MIRAUS achieves superior segmentation performance across the evaluated settings. Standard fully automatic single-modality architectures (Rows 1–4) exhibit limited spatial accuracy due to standard TRUS contrast constraints, with the top performer (nnU-Net) achieving 84.45% Dice. Crucially, despite being trained on paired MRI-TRUS data and specifically engineered to accommodate missing modalities during inference, architectures such as RFNet and mmFormer (Rows 5–6) still yield suboptimal TRUS-only results (Dice of 76.46% and 69.94%). This indicates that simply training multi-modal architectures to tolerate missing modalities remains insufficient for prostate segmentation, motivating more explicit transfer of training-time anatomical information.

By framing cross-modal learning through privileged information distillation (LUPi), MIRAUS (automatic, Row 8) avoids any multi-modal inference dependency altogether. It significantly outperforms existing feature-prototypical distillation strategies (ProtoKD in Row 7, 79.99% Dice), advancing automatic performance to 93.20% Dice, 87.41%

IoU, and reducing HD95 boundary error to 3.62 mm. Furthermore, when evaluated against interactive, box-prompted foundation models (Rows 9–13), MIRAUS (box-assisted, Row 14) achieves competitive internal accuracy (92.56% Dice) and the highest external box-assisted Dice among the evaluated methods (84.72%). Notably, even in its automatic full-image-prompt mode, MIRAUS (93.20% Dice) surpasses all box-assisted foundation models on the internal cohort (e.g., SAMUS box-only at 92.07% and LiteMedSAM at 88.82%), indicating that the MRI-privileged training strategy can provide complementary anatomical information beyond coarse localization prompts.

To visually inspect volumetric shape and slice-level boundary precision, Fig. 3 compares MIRAUS with representative automatic (nnU-Net) and prompt-guided (LiteMedSAM) baselines on the internal cohort across the apex, mid-gland, and base. In the illustrated case, nnU-Net shows substantial shape distortion and boundary over-estimation where the prostate boundary blends into adjacent structures. LiteMedSAM remains spatially localized by its box prompt, but exhibits local boundary deviations across the gland, including mild contour displacement at the apex, pronounced over-segmentation around the mid-gland, and outward boundary expansion at the base. In contrast, the MIRAUS contours more closely follow the reference annotations across all three anatomical zones, preserving both the global gland shape and local boundary geometry. Extended visual comparisons across 15 anonymized cases are available in our online result viewer (<https://zeming-huang.github.io/MIRAUS/>).

On the separate-center external cohort (China–Japan Friendship Hospital dataset), fully automatic MIRAUS achieved 71.30% Dice, outperforming the evaluated non-prompted automatic baselines. Under the standardized reference-mask-derived box condition shared by all box-prompted methods, MIRAUS achieved 84.72% Dice, 73.91% IoU, and an HD95 boundary error of 32.05 pixels. This 13.42-point Dice boost underscores the value of human-in-the-loop interaction for cross-center deployment. Minimal user prompting provides critical spatial anchoring to mitigate localized imaging artifacts and scale variations, while introducing only modest manual burden.

As illustrated by the qualitative slice comparisons in Fig. 4, the fully automatic nnU-Net baseline (purple contours) shows substantial degradation under the external-domain shift, with pronounced under-segmentation and locally displaced boundaries across the three cases. The box-prompted LiteMedSAM baseline (blue contours) remains substantially more robust in terms of overall prostate localization, but noticeable local boundary deviations persist, including regional over-extension, inward displacement, and imperfect tracking of irregular contour segments. In contrast, MIRAUS (red contours) more closely follows the reference boundary at these locally challenging regions, better preserving asymmetric protrusions, concave segments, and other patient-specific shape variations. Importantly, the figure intentionally includes challenging cases in which MIRAUS does not uniformly outperform the prompted baseline in

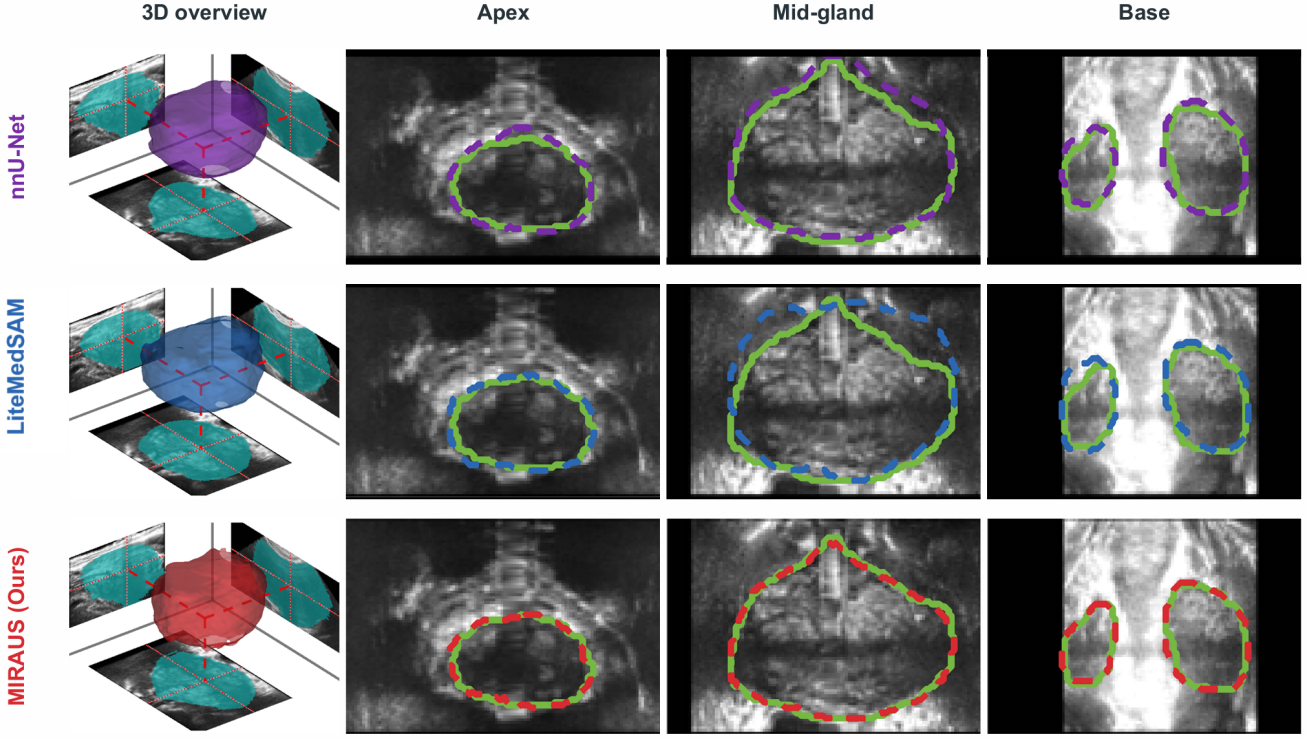


Figure 3: Qualitative 3D volumetric and 2D multi-slice segmentation comparison on the public μ RegPro dataset across Apex, Mid-gland, and Base regions. From top to bottom: nnU-Net (purple), LiteMedSAM (blue), and the proposed MIRAUS (red), with ground-truth contours delineated in solid green.

Table 2

Segmentation performance under five-fold cross-validation and external cohort

Method	Params (M)	μ RegPro			External cohort		
		Dice (%) $\uparrow$	IoU (%) $\uparrow$	HD95 (mm) $\downarrow$	Dice (%) $\uparrow$	IoU (%) $\uparrow$	HD95 (pixel) $\downarrow$
<i>Fully automatic methods (TRUS-only)</i>							
1 3D U-Net [23]	4.81	77.39	66.78	10.67	—	—	—
2 UNETR++ [50]	42.95	77.52	67.21	11.56	—	—	—
3 nnU-Net [24]	24.27	84.45	75.95	8.85	52.72	39.95	83.76
4 SAMed [32]	91.35	75.07	65.76	37.10	66.53	54.22	39.18
5 RFNet [44]	8.69	76.46	64.68	36.63	—	—	—
6 mmFormer [45]	36.65	69.94	56.67	72.74	—	—	—
<i>Privileged distillation methods (MRI-guided training, TRUS-only inference)</i>							
7 ProtoKD [46]	3.27	79.99	70.04	23.14	—	—	—
8 MIRAUS (automatic)	15.92	93.20	87.41	3.62	71.30	57.25	59.20
<i>Box-assisted methods (Interactive / Box-prompted inference)</i>							
9 LiteMedSAM [11]	9.79	88.82	80.29	3.79	83.75	72.28	34.47
10 SAM-Med2D [33]	271.24	85.78	76.82	13.66	44.43	32.14	99.92
11 SegVol [36]	180.89	82.79	73.81	17.89	—	—	—
12 nnInteractive [37]	383.50	78.32	65.37	25.73	76.99	63.97	62.30
13 SAMUS [34]	133.32	92.07	85.71	8.33	—	—	—
14 MIRAUS (box-assisted)	15.92	92.56	86.36	4.19	84.72	73.91	32.05

Note: Missing entries (—) indicate methods not evaluated on the external cohort under the corresponding experimental setting; such methods rely on volumetric or multimodal inputs that are unavailable in the external 2D TRUS cohort. External HD95 values are reported in pixels because the exported images lack validated physical pixel-spacing metadata.

Dice overlap. Even in these cases, similar global overlap can correspond to different local contour fidelity, with MIRAUS more closely matching the reference geometry along difficult boundary segments. Together, these examples illustrate both

the robustness and the limitations of MIRAUS under cross-center domain shift, while suggesting that MRI-privileged training can help preserve local anatomical boundary detail beyond coarse spatial localization alone.

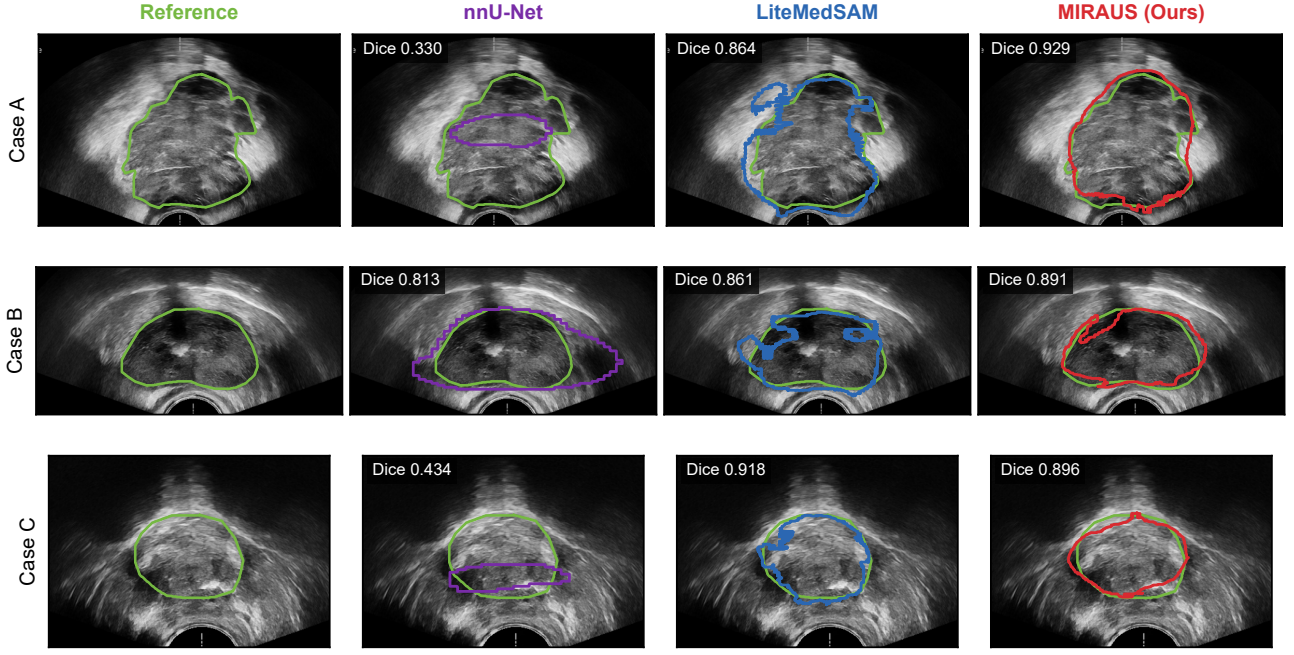


Figure 4: Qualitative comparison of segmentation performance on the external verification cohort (China–Japan Friendship Hospital dataset).

4.2. Ablation Study

We systematically validate the core methodological components of MIRAUS: utility-supervised multi-slice MRI aggregation in the teacher model, and the residual privileged distillation scheme in the student model. Table 3 summarizes the complete ablation hierarchy evaluated in TRUS-only deployment mode alongside the fine-tuned LiteMedSAM baseline.

Efficacy of Utility-Supervised MRI Aggregation Table 3 evaluates teacher aggregation strategies against the fine-tuned LiteMedSAM baseline (88.82% Dice, 6.62 mL Volume MAE).

Even unguided aggregation strategies (A1 and A2) achieve substantial performance gains over the foundation baseline (improving Dice to 92.07% and 91.36%, and reducing Volume MAE to 4.41 mL and 4.84 mL, respectively). This baseline gain confirms that aggregating multi-slice privileged MRI context in the teacher model is beneficial for cross-modal transfer. However, unguided feature fusion remains sub-optimal because it does not explicitly condition the teacher model’s representations on downstream student transferability. In contrast, introducing explicit segmentation utility supervision (A3) encourages the teacher model to form boundary-relevant representations from the local MRI neighborhood. This targeted guidance yields further comprehensive improvements across all evaluation metrics, raising the student model’s Dice score to 93.20%, reducing HD95 to 3.62 mm, and advancing volume estimation precision to 3.61 mL Volume MAE—achieving a reduction of over 45% in volumetric error over fine-tuned LiteMedSAM.

Efficacy of Residual Privileged Distillation Regarding the student distillation objective using the proposed utility-guided teacher model (A3), conventional logit distillation B2 proves effective over the non-distilled TRUS baseline B1 (91.94% Dice, 4.22 mL Volume MAE), advancing performance to 92.86% Dice and 3.64 mL Volume MAE. This demonstrates that transferring overall output probability distributions helps regularize the lightweight student model.

Crucially, our proposed residual privileged distillation objective B3 outperforms conventional KD across all metrics, reaching the overall top performance of 93.20% Dice, 87.41% IoU, 3.62 mm HD95, and 3.61 mL Volume MAE. While conventional KD only transfers coarse output logit distributions, B3 directly supervises the student model to predict the fine-grained incremental anatomical representation $\mathbf{r}_T$ provided by privileged MRI. This formulation enables the lightweight TRUS student model to effectively internalize cross-modal structural priors and achieve maximal performance (gains of 4.38 percentage points in Dice and 7.12 percentage points in IoU over fine-tuned LiteMedSAM).

4.3. Analysis of Privileged MRI Context and Modality Dependence

To understand how privileged multi-slice MRI representations empower cross-modal distillation, we evaluated the effect of training-time MRI context size and verified the teacher model’s functional dependence on MRI input.

Impact of Privileged MRI Neighborhood Size During privileged training, the teacher model aggregates a spatial neighborhood of multi-slice MRI representations to construct

Table 3

Ablation study in TRUS-only deployment mode.

ID	Teacher Agg.	Student KD	Dice (%) $\uparrow$	IoU (%) $\uparrow$	HD95 (mm) $\downarrow$	Vol MAE (mL) $\downarrow$
–	–	LiteMedSAM (Fine-tuned)	88.82 $\pm$ 5.41	80.29 $\pm$ 7.92	3.79 $\pm$ 2.15	6.62 $\pm$ 6.10
A1	Uniform Avg.	Residual ($\mathcal{L}_{priv}$)	92.07 $\pm$ 3.89	85.53 $\pm$ 6.35	4.44 $\pm$ 2.07	4.41 $\pm$ 4.39
A2	Unguided Soft	Residual ($\mathcal{L}_{priv}$)	91.36 $\pm$ 4.83	84.43 $\pm$ 7.66	4.55 $\pm$ 2.42	4.84 $\pm$ 5.34
A3	Util.-Supervised	Residual ($\mathcal{L}_{priv}$)	93.20 $\pm$ 3.00	87.41 $\pm$ 5.05	3.62 $\pm$ 1.81	3.61 $\pm$ 4.00
B1	Util.-Supervised	None (TRUS-only)	91.94 $\pm$ 5.20	85.45 $\pm$ 7.84	4.26 $\pm$ 2.23	4.22 $\pm$ 5.47
B2	Util.-Supervised	Logit KD	92.86 $\pm$ 3.37	86.84 $\pm$ 5.60	3.70 $\pm$ 1.86	3.64 $\pm$ 3.79
B3	Util.-Supervised	Residual ($\mathcal{L}_{priv}$)	93.20 $\pm$ 3.00	87.41 $\pm$ 5.05	3.62 $\pm$ 1.81	3.61 $\pm$ 4.00

rich anatomical context for the TRUS-only student model. As detailed in Table 4(a), expanding the training-time MRI context from a single slice to 3 and 5 slices progressively enhances student inference performance, with the 5-slice configuration achieving the highest accuracy (93.20% Dice and 87.41% IoU). However, further expanding the context window to 7 slices leads to a marked performance degradation (87.60% Dice, representing a 5.60 pp drop from the reference). This indicates that an excessively wide spatial window introduces weakly correlated out-of-plane features that dilute essential local anatomical guidance. Thus, a compact 5-slice MRI neighborhood provides the best-performing context size among the evaluated configurations.

Counterfactual Verification of Teacher MRI Dependence To examine the teacher network’s dependency on privileged MRI cues rather than functioning as a redundant feature extractor, we conducted counterfactual input interventions across TRUS frames (Table 4(b)). Under standard matched privileged MRI input, the frozen teacher model achieves a baseline segmentation performance of 86.40% Dice.

When evaluating input sensitivity, completely zeroing out the MRI feature representation serves as an out-of-distribution intervention on the frozen teacher model, causing a performance degradation of 21.59 percentage points. This substantial decrease indicates that the privileged teacher model relies heavily on cross-modal representations to construct its residual supervision targets. Furthermore, replacing the patient-matched MRI with cross-patient scans results in a moderate performance decrease of 2.69 percentage points. This behavior suggests that the privileged branch utilizes two complementary components: the drop under cross-patient mismatch reflects the omission of subject-specific anatomical boundary cues, whereas the residual 83.70% Dice score demonstrates that the architecture still extracts transferable, general prostate structural priors from unmatched MRI slices.

5. Discussion

5.1. Mechanistic Insights into Privileged Feature Distillation

An important question for interpreting MIRAUS is how anatomical information available only during training

Table 4

Analysis of Privileged Teacher MRI Utilization

(a) Effect of Privileged MRI Context Size			
MRI Slices	Dice (%) $\uparrow$	Δ Dice (pp)	IoU (%) $\uparrow$
1 Slice	90.84	−2.36	83.33
3 Slices	92.15	−1.05	85.55
5 Slices	93.20	Reference	87.41
7 Slices	87.60	−5.60	78.39

(b) Counterfactual Teacher Dependence on Privileged MRI Input			
MRI Condition	Dice (%) $\uparrow$	Δ Dice (pp)	SD
Matched MRI	86.40	Reference	$\pm$ 3.94
Cross-Patient Mismatch	83.70	−2.69	$\pm$ 4.10
Zeroed MRI Input	64.81	−21.59	$\pm$ 12.70

remains useful after the privileged MRI branch is removed. This question is particularly relevant because the MRI input is only coarsely related to the corresponding TRUS frame, while the deployed student model must ultimately make its measurement from TRUS alone. The central issue is therefore not simply whether MRI improves training, but how the privileged information is filtered, transferred, and reflected in the behavior of the resulting TRUS model.

The preceding ablation and modality-dependence analyses provide the first part of this picture by showing that the teacher model makes functional use of local MRI context and that patient-matched MRI contributes additional information beyond generic cross-modal input. The remaining question is how this privileged information manifests after transfer to the student model. We examine this at two complementary levels: first, at the prediction level, by inspecting how the student model modifies local boundary confidence; and second, at the representation level, by testing whether the learned student correction becomes more aligned with the MRI-conditioned residual target. These two views are then connected to the measurement-oriented deployment analyses that follow.

At the prediction level, Fig. 5 compares the local confidence patterns of MIRAUS and LiteMedSAM on a representative case. As shown in Fig. 5(b), MIRAUS achieves a higher local Dice score (0.969 vs. 0.878) and produces a probability boundary that more closely follows the reference contour. The difference map in Fig. 5(c) further localizes where the two predictions diverge. Along the lateral prostate boundary,

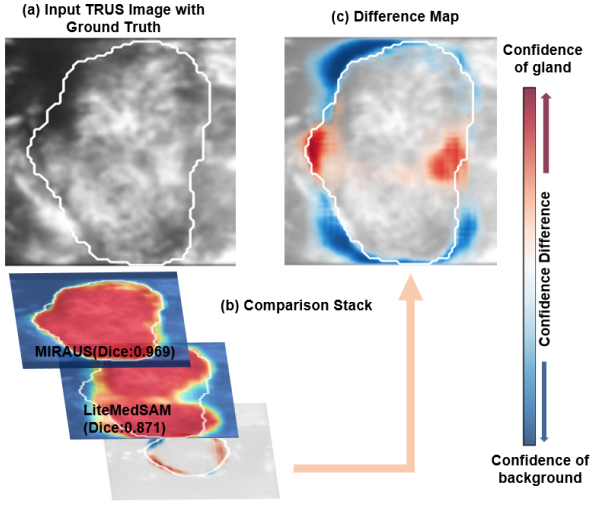


Figure 5: Slice-confidence difference analysis between MIRAUS and fine-tuned LiteMedSAM. (a) Input TRUS frame overlaid with ground truth (white contour). (b) Predicted probability maps for MIRAUS and LiteMedSAM. (c) Difference map showing region-wise confidence divergence (red: higher MIRAUS confidence for prostate tissue; blue: higher confidence for background).

MIRAUS shows higher gland confidence in regions that are under-segmented by LiteMedSAM, while near the peripheral border it assigns higher background confidence in areas affected by boundary leakage in the LiteMedSAM prediction. These changes reduce both missed gland regions and spurious peripheral responses in the illustrated case, resulting in a cleaner and more sharply localized contour. These localized changes are consistent with the broader reduction in boundary error observed across the cohort and illustrate how the transferred privileged information can sharpen local boundary discrimination in the TRUS student model.

The prediction-level differences alone do not establish whether the transferred privileged information is also reflected in the student model’s internal representation. We therefore analyzed the feature space using debiased linear Centered Kernel Alignment (CKA) [51] (Fig. 6). The analysis included the TRUS base feature F_U , the MRI feature F_M , the teacher-defined increment r_T , and the student corrections learned without distillation (Δ_{B1}), with logit distillation (Δ_{B2}), and with residual distillation (Δ_{B3}). CKA was computed independently within each cross-validation fold and then averaged across the five folds.

The mean CKA matrix shows that F_M remains representationally distinct from F_U (0.07) and from the student corrections (0.10–0.13). More importantly, the correction learned with residual distillation (Δ_{B3}) shows greater alignment with the teacher-defined target r_T (CKA = 0.47) than either the non-distilled (0.36) or logit-distilled (0.37) alternatives, with the same direction of change observed across all five folds (mean gain of +0.108). At the same time, its alignment with the original TRUS feature F_U decreases from 0.64 to

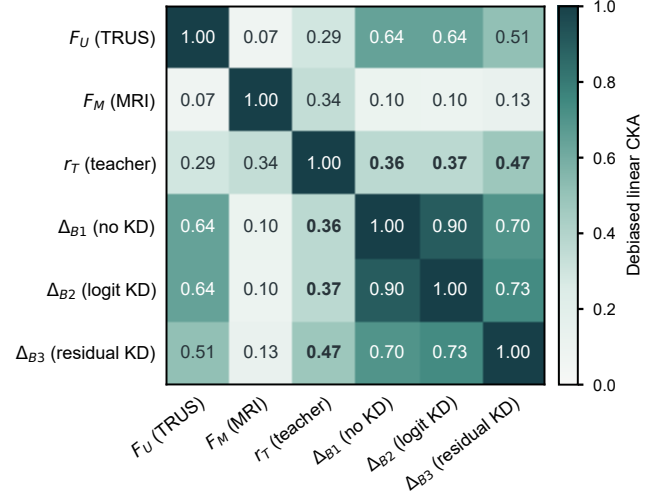


Figure 6: Centered Kernel Alignment (CKA) similarity matrix across internal representations. Mean CKA scores evaluate feature-space alignment between single-modality sources (F_U , F_M), the teacher-defined residual target (r_T), and student-learned task corrections without distillation (Δ_{B1}), with logit distillation (Δ_{B2}), and with residual distillation (Δ_{B3}).

0.51. This coordinated shift is consistent with the intended role of residual distillation: the student correction becomes more closely aligned with the MRI-conditioned increment rather than remaining dominated by the original TRUS representation. Together with the confidence-map analysis, these findings connect the transferred privileged information to both the student model’s local boundary behavior and its internal feature adaptation.

5.2. Deployment Characteristics and Measurement Analysis

The preceding analyses describe how privileged MRI information is reflected in the student model’s predictions and internal representation. A separate question is whether these effects remain meaningful under the practical conditions of TRUS measurement. We therefore examined four deployment-relevant dimensions: computational efficiency, anatomical depth sensitivity, robustness to operator-provided localization, and agreement with conventional caliper-based volume measurement.

Model Complexity and Runtime Latency As summarized in Table 2, MIRAUS balances segmentation performance and computational footprint. With 15.92 M parameters, MIRAUS is lighter than foundation models such as SAM-Med2D (271.24 M), SegVol (180.89 M), and nnInteractive (383.50 M). The TRUS-only student model requires no privileged MRI processing at deployment. Inference on a single 2D TRUS frame takes under 15 ms on an NVIDIA GeForce RTX 5070 Ti GPU. In box-assisted mode, manual box placement required 4.12–5.60 s per frame across observers, and the complete measured workflow, including inference and post-processing, required 4.36–5.85 s per frame.

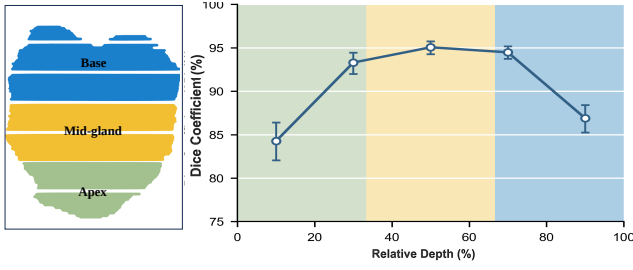


Figure 7: Segmentation performance across relative axial prostate depths. (Left) Anatomical region categorization across five equal depth intervals. (Right) Patient-macro mean regional 3D Dice scores (%) across relative axial depths. Error bars indicate 95% confidence intervals obtained from 20,000 patient-level bootstrap resamples.

These retrospective measurements indicate that model-side inference contributes little to the overall interaction time, which is dominated by manual localization in the box-assisted workflow.

Anatomical Zonal Sensitivity along Axial Depth To examine depth-dependent segmentation performance across anatomical sub-regions, each patient’s gland-bearing slice sequence was normalized to a relative axial depth of 0%–100% and categorized into five equal intervals (Fig. 7). Within each interval, regional 3D Dice was calculated per patient by pooling foreground counts across constituent slices and averaged across patients with equal weighting. Uncertainty in the estimated mean performance was quantified using 95% confidence intervals derived from 20,000 patient-level bootstrap resamples. As shown in Fig. 7, MIRAUS achieves its peak accuracy in the central depth interval (40%–60%, mean regional Dice 95.06%; 95% CI: 94.27%–95.75%). Performance naturally declines toward the terminal apical (0%–20%, 84.27%) and basal (80%–100%, 86.89%) intervals. This pattern aligns with clinical expectations, as mid-gland slices offer well-defined anatomical contours, whereas apical and basal regions suffer from severe acoustic shadowing, low signal-to-noise ratios, and ill-defined prostate-bladder/rectal interfaces. Nevertheless, as qualitative comparisons confirm (Fig. 3), privileged MRI distillation significantly mitigates boundary degradation in these challenging terminal slices, avoiding the severe boundary collapse common in non-privileged baselines.

Operator-Assisted Localization and Robustness to Box Imperfections On the separate-center TRUS cohort, fully automatic inference showed performance degradation under variations in field of view, imaging gain, and scale. We therefore evaluated box-assisted inference as a complementary operating mode. To quantify robustness to variable operator-supplied bounding boxes, ground-truth segmentations annotated by a senior urological imaging expert were adopted as the reference standard. Two clinical observers with differing levels of experience independently placed bounding

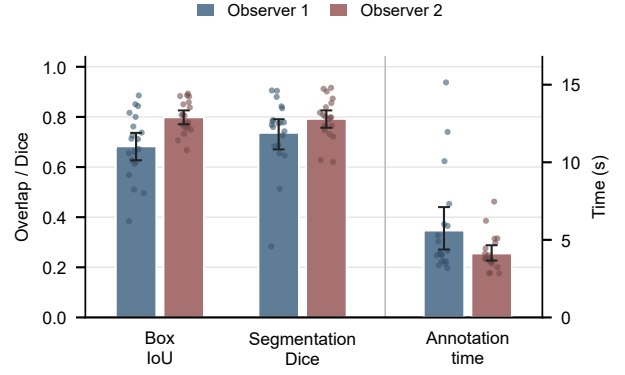


Figure 8: Quantitative comparison of inter-operator variability during box-assisted deployment across two observers with differing clinical experience. Metrics evaluate Box IoU against expert reference, resulting segmentation Dice, and annotation time.

boxes on a representative evaluation subset of external-center TRUS frames without viewing reference contours. Inter-observer box positioning achieved a mean overlap IoU of 0.736 ± 0.134 . As detailed in Fig. 8, Observer 2 (more experienced) placed more accurate bounding boxes relative to the expert reference (0.798 ± 0.064 vs. 0.683 ± 0.130 Box IoU), which correspondingly led to higher downstream segmentation performance (0.792 ± 0.081 vs. 0.736 ± 0.142 Dice). In terms of operational efficiency, Observer 2 required less annotation time per frame (4.1 ± 1.2 s vs. 5.6 ± 3.2 s). Notably, even under the less accurate prompts from Observer 1, MIRAUS maintained robust segmentation output (0.736 Dice).

Given the intrinsic anatomical ambiguity and severe boundary noise in prostate TRUS—where inter-expert human agreement itself inherently exhibits noticeable variation [22]—these results highlight the practical resilience of MIRAUS. Even with coarse, operator-dependent bounding boxes, the framework delivers stable segmentation performance while maintaining a rapid, clinically practical annotation workflow.

Ellipsoid Volume Measurement Integration To assess how MIRAUS supports prostate volume measurement under both direct volumetric and conventional caliper-based workflows, we compared three estimation strategies against the expert manual 3D segmentation reference (V_{ref}) across 73 patients (Fig. 9): (i) **Direct 3D segmentation** ($V_{\text{model}}^{\text{3D}}$), obtained by voxel-wise integration of the predicted 3D prostate mask; (ii) **Model-assisted ellipsoid** ($V_{\text{model}}^{\text{ellip}}$), calculated using Eq. (2) with model-derived transverse and anteroposterior diameters, the reference longitudinal diameter, and a GT-selected axial measurement frame; and (iii) **GT-mask-derived ellipsoid** ($V_{\text{GT}}^{\text{ellip}}$), calculated using the same ellipsoid formulation with standardized calipers extracted from the expert ground-truth masks.

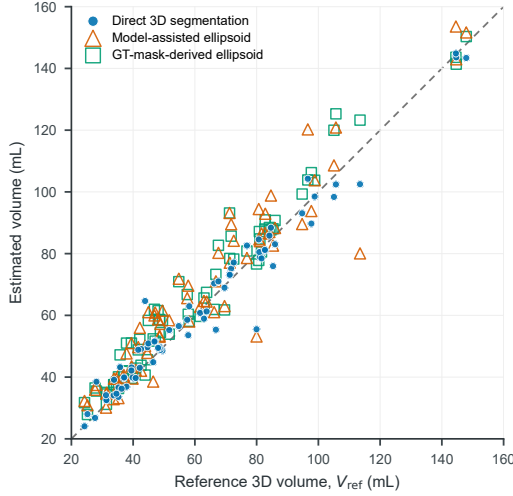


Figure 9: Volumetric estimation accuracy and clinical workflow integration across 73 patients. Comparison among automated 3D direct segmentation (V_{model}^{3D}), model-assisted ellipsoid estimation ($V_{\text{model}}^{\text{ellip}}$), and GT-mask-derived ellipsoid estimation ($V_{\text{GT}}^{\text{ellip}}$) against the expert manual 3D segmentation gold standard (V_{ref}).

As shown in Fig. 9, direct 3D segmentation (blue filled circles) provides the most accurate volume estimates, with most measurements closely distributed around the identity line. Relative to V_{ref} , it achieved a bias of only +0.60 mL and an MAE of 3.61 mL, demonstrating that the predicted prostate boundaries support accurate direct volumetric measurement.

We next examined compatibility with the conventional caliper-based workflow. Model-assisted ellipsoid estimates (orange open triangles) achieved a bias of +4.80 mL, an MAE of 7.50 mL, and a MAPE of 13.43% relative to V_{ref} . The GT-mask-derived ellipsoid estimates (green open squares) showed a similar systematic overestimation, with a bias of +5.48 mL, an MAE of 6.28 mL, and a MAPE of 11.82%. The comparable positive bias of the two ellipsoid-based strategies indicates that a substantial part of the discrepancy relative to the 3D reference arises from the geometric approximation inherent to the conventional ellipsoid formula rather than from model-derived boundary measurement alone.

To isolate the additional effect of replacing reference in-plane calipers with model-derived measurements, we directly compared $V_{\text{model}}^{\text{ellip}}$ with $V_{\text{GT}}^{\text{ellip}}$. The resulting signed difference was -0.68 mL, with an MAE of 4.68 mL. Because the reference superior-inferior diameter and the ellipsoid formulation are shared between the two estimates, this comparison specifically reflects the volumetric effect of substituting the transverse and anteroposterior reference calipers with measurements derived from MIRAUS boundaries. Although errors in the two in-plane diameters propagate multiplicatively through the ellipsoid formula, the resulting additional discrepancy remained modest relative to the overall ellipsoid approximation error.

These results support two complementary measurement modes. Direct 3D integration provides the highest volumetric accuracy when complete predicted masks are available, whereas model-derived calipers remain compatible with the conventional ellipsoid reporting workflow. Together with sub-15 ms inference, this enables MIRAUS to support both automated volumetric measurement and rapid caliper-based assessment while reducing dependence on manual in-plane caliper placement.

5.3. Limitations and Future Directions

While MIRAUS demonstrates strong efficacy and clinical utility, several methodological scope boundaries highlight promising avenues for future research.

First, regarding anatomical and diagnostic granularity, the current framework primarily focuses on whole-gland prostate segmentation. In advanced targeted biopsy or focal therapy, delineating intra-prostatic malignant lesions, for example PI-RADS ≥ 3 targets, and managing severe tissue deformation from benign prostatic hyperplasia (BPH) remain critical. A crucial next step is to extend the privileged learning formulation to multi-task architectures that simultaneously distill spatial gland boundaries and focal lesion priors from multi-parametric MRI.

Second, the present volume and operator analyses are retrospective, and the external cohort is assessed at the 2D frame level. Although these findings establish technical segmentation and measurement performance across acquisition conditions, their downstream clinical impact, including associations with PSA density, biopsy yield, focal treatment planning, and real-time workflow efficiency, remains to be examined. Prospective multi-center studies across diverse ultrasound vendors will be essential to translate these technical gains into measurable clinical utility.

Third, regarding multi-slice spatial consistency, processing independent 2D frames offers distinct speed advantages for real-world deployment, yet it does not explicitly enforce inter-slice contour continuity. When extending the model to continuous multi-planar sweeps or 3D volume reconstruction, frame-by-frame inference may introduce inter-slice jitter or boundary fluctuations. Future extensions will explore 3D temporal-spatial smoothing, volumetric tracking, or joint multi-slice attention to guarantee continuous 3D spatial coherence during dynamic scanning.

6. Conclusion

We proposed MIRAUS, an MRI-informed residual adaptation framework for TRUS prostate boundary measurement under learning using privileged information. MRI is used only during training to provide complementary anatomical context, while deployment remains TRUS-only. To accommodate the coarse correspondence between independently acquired MRI and TRUS, MIRAUS combines local 2.5D MRI context, utility-supervised aggregation, and residual transfer of MRI-conditioned feature increments to a lightweight TRUS student model. The deployed model supports both automatic and box-assisted boundary measurement without requiring

MRI. In patient-disjoint five-fold cross-validation on 73 μ RegPro patients, automatic inference achieved 93.20% Dice and 3.62 mm HD95. In the retrospective caliper workflow, model-derived in-plane measurements produced a 4.68-mL MAE in ellipsoid volume estimation relative to GT-derived calipers. On the separate-center cohort, box-assisted inference achieved 84.72% Dice. Together, these results show that training-time MRI anatomy can improve TRUS-only boundary and caliper measurement without becoming a deployment requirement.

CRedit authorship contribution statement

Zeming Huang: Conceptualization, Methodology, Software, Formal analysis, Validation, Visualization, Writing - original draft. **Haixin Chen:** Investigation, Resources, Data curation, Validation, Writing - review and editing. **Chuanzhen Cao:** Investigation, Resources, Data curation, Writing - review and editing. **Tao Wang:** Conceptualization, Methodology, Supervision, Project administration, Funding acquisition, Writing - review and editing. **Zhipei Huang:** Methodology, Validation, Writing - review and editing. **Fei Qin:** Supervision, Project administration, Writing - review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The public μ RegPro dataset used in this study is available from Zenodo (version 1.1.0; <https://doi.org/10.5281/zenodo.8004388>). The self-collected TRUS cohort contains human-participant imaging data and cannot be made publicly available because the participant consent and institutional ethics approval do not permit unrestricted distribution. Method-verification source code and the five-fold patient assignments are publicly available at <https://github.com/Zeming-Huang/MIRAUS>. Representative predictions and visualizations are available at <https://zeming-huang.github.io/MIRAUS/>.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this manuscript, Chatgpt was used to assist with language editing. The authors reviewed and edited the assisted text and take full responsibility for the content of the manuscript.

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